From Gaussian Samples to a Complex Wishart Minimum Law in Lean

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Abstract

We present a Lean formalization of selected complex Gaussian Gram matrix laws, with a complete actual-model chain for the smallest eigenvalue of a central complex 2×2 Wishart matrix. The construction starts from independent real standard Gaussian coordinates, fixes the unit complex variance convention, and derives an independent Bartlett coordinate law. An explicit four-real-coordinate chart yields the normalized Gram density on the positive definite cone. Translation of this cone, followed by a Hermitian spectral threshold criterion, identifies the actual minimum law as exponential with rate two. We also formalize its positive scalar variance dependence, with a log-rate outage expression as a monotone-event corollary. Supporting developments include noncentral one-column results, an arbitrary-row one-column small-threshold asymptotic, and fixed-subspace Gaussian projection and independence interfaces. We distinguish these actual probability laws from general-dimensional candidate-kernel identities and unresolved joint spectral densities. The mathematical distribution formulas are classical; the contribution is their explicit formal connection to an underlying Gaussian sample model and the reusable interfaces needed for that connection. We also document the AI-assisted proof workflow and its limits: one recorded session contained 38 local-prover generations, of which two guided helper proofs passed the kernel, while five separate adversarial-review requests yielded no usable review. These observational records and a declaration-level provenance map accompany the proofs; they are not a controlled model benchmark. A subsequent scoped cross-family model review reported no concrete mathematical errors; human mathematical review remains open.

1 Introduction

Wishart eigenvalues occur in multivariate probability and in the analysis of multiple-antenna channels. In eigenmode transmission, an eigenvalue determines the gain of a scalar subchannel. Jin, McKay, Gao, and Collings [1] give formulas for ordered eigenvalue distributions of complex noncentral Wishart matrices and apply them to communication performance. This paper studies the formal verification of selected probability statements underlying that analysis.

The main difficulty is connecting representations. A Gaussian sample can be represented by real coordinates, complex columns, a Gram matrix, coordinates on the Hermitian space, and ordered eigenvalues. A density identity at one level does not automatically establish the law at another. Variance factors, the reference measure, exceptional sets, and strict versus weak events must be tracked through each transition. We organize the development around equalities of pushforward measures so that a proposed density cannot enter as an assumed description of the model it is meant to identify.

Our principal example is a central complex matrix $H \in \mathbb{C}^{2\times 2}$ with independent unit-variance entries. We prove the coordinate density of H^*H and derive

$$\mathbb{P}\{\lambda_{\min}(H^*H) > x\} = e^{-2x}, \qquad x \geq 0.$$

The proof uses a translation of the positive definite cone and does not require an ordered spectral Jacobian. Positive scalar covariance and a logarithmic-rate statistic are then obtained by transporting events through this actual law.

The contribution has three parts. First, the central 2×2 chain exposes the distributional, coordinate, and spectral obligations separately. Second, the project specializes existing Gaussian tools to the sample encoding and Hermitian coordinates, together with a dimension-independent noncentral one-column asymptotic. Third, it supplies a source-linked account of the AI-assisted workflow, proof-body provenance, accepted and rejected attempts, and the remaining review obligations. We make no claim that the classical distribution formulas are new, or that this is the first formalization of Wishart theory. A comprehensive novelty comparison remains part of release preparation.

The development builds on Lean and mathlib [9]. In particular, the change-of-variables infrastructure developed in mathlib [10] and its Gaussian and independence libraries carry much of the foundational analysis. The role of probability kernels in this ecosystem is described by Degenne [11]. Our contribution specializes and connects those interfaces to concrete Gaussian matrix statistics; it does not replace their underlying general theorems.

1.1 Related mathematical and formal work

Bartlett's independent-factor treatment of regression is a classical source for the decomposition viewpoint [3]; Goodman developed multivariate complex Gaussian analysis [4]. These precede the present formalization. Edelman's work treats exact Wishart extreme-eigenvalue laws [5, 6]; complex Gaussian scaling conventions must be converted before comparing constants. We use unit complex variance as in (1), rather than assigning variance one to each real component.

Probability formalization also predates this project in other systems. Avigad, Hölzl, and Serafin formalized the central limit theorem in Isabelle/HOL [7]; Affeldt and Cohen developed measure construction and integration in Coq/Mathematical Components [8]. These are foundational comparisons, not claims that either development contains the Wishart result studied here. Our targeted search does not establish the absence of a prior Wishart formalization, and no cross-system priority claim is made.

Within mathlib, joint Gaussianity, zero cross covariance implying independence, and the product-law characterization of independent maps are existing results. Our orthogonal-projection statements below are wrappers and specializations of those results. The concrete sample encoding, its Bartlett-to-Gram chart, the chosen Hermitian coordinate measure, and their composition into an actual minimum law are the integration work reported here. We do not count general Gaussian independence as a new contribution.

2 Model, measures, and formal scope

Let U_{ij}, V_{ij} be independent $N(0,1)$ variables and set

$$H_{ij} = \frac{U_{ij} + \mathrm{i}V_{ij}}{\sqrt{2}}, \qquad X = M + H, \qquad W = X^*X. \tag{1}$$

Thus $\mathbb{E}|H_{ij}|^2 = 1$. The complex scalar density is $\pi^{-1}\exp(-|z|^2)$ relative to $d\Re z\, d\Im z$. The sample space in Lean is a finite-dimensional real Euclidean space named `ComplexSample`; `complexSampleMatrix` performs the complex encoding. The underlying law is mathlib's `stdGaussian`. The deterministic mean M is added after encoding. The Gram law is a measurable pushforward, rather than a measure defined initially by a Wishart density formula.

For a general $t \times s$ sample we use X^*X when $s \le t$. The Lean definition `complexNoncentralSampleSmallestEigenvalue` is broader: for an $m \times n$ sample with $m < n$, the $n \times n$ matrix X^*X is singular and its smallest eigenvalue is identically zero. Thus that definition alone does not denote the smallest *nonzero* eigenvalue in the wide case. The actual laws in this manuscript use the tall or square convention $s \le t$. For a one-column sample, the only Gram eigenvalue is $\|X\|_F^2$. For the main example, $s = t = 2$ and $M = 0$. Eigenvalues are ordered decreasingly; the last coordinate is the minimum. The formal real-valued statistic `twoRowActualMinimum` is computed from the actual sample Gram matrix.

2.1 The Hermitian reference measure

Write

$$W = \begin{pmatrix} u & b + \mathrm{i}c \\ b - \mathrm{i}c & v \end{pmatrix}. \tag{2}$$

The reference measure is $du\,db\,dc\,dv$. The positive definite coordinate cone is

$$D = \{(u, b, c, v) : u > 0,\ b^2 + c^2 < uv\}.$$

It is a four-dimensional real coordinate measure on Hermitian matrices. The Hermitian subspace is null for eight-dimensional ambient Lebesgue measure; restricting that ambient measure to it gives the zero measure. The nonzero Euclidean measure induced *on* the subspace by the real Frobenius metric is another convention. In (2),

$$ds_F^2 = du^2 + dv^2 + 2\,db^2 + 2\,dc^2, \qquad d\,\mathrm{vol}_F = 2\,du\,db\,dc\,dv.$$

Consequently (5) has coefficient π^{-1} relative to our coordinate measure, and $(2\pi)^{-1}$ relative to induced Frobenius volume. This normalization comparison is kept distinct from the actual-law proof. The module `HermitianCoordinateMetric` checks the coordinate energy identity in arbitrary dimension. In size two it exhibits the inverse frame $\mathrm{diag}(1, 1/\sqrt{2}, 1/\sqrt{2}, 1)$, proves that pushing coordinate Lebesgue measure through this map multiplies it by two, and re-expresses the actual Gram law with coefficient $(2\pi)^{-1}$ relative to that frame measure. The exact entry point is `actualTwoRow_Gram_frobeniusFrameDensity`. These statements specify the frame explicitly; identification with an intrinsically defined subspace volume remains a semantic review obligation.

In dimension s, `CentralHermitianCoordinates` constructs an explicit measurable equivalence between s^2 real coordinates and the Hermitian subspace. Its coordinate measure is a product of real Lebesgue measures, transported by this equivalence. Each unordered off-diagonal pair contributes two real directions of Frobenius squared length two. Thus in dimension s, induced Frobenius volume is $2^{s(s-1)/2}$ times this coordinate measure, and a density coefficient relative to Frobenius volume is divided by that factor. This general volume factor is derived here geometrically; we do not claim a general-dimensional Lean volume theorem. This construction fixes a normalization; it does not by itself identify any Gaussian Gram density.

2.2 Meaning of the verification claims

An actual-law theorem starts with the sample measure in (1) and identifies its pushforward. A coordinate theorem gives an equivalence, a Jacobian, or a transformation of integrals. A candidate identity simplifies a proposed analytical expression. These are recorded separately. In particular, a theorem about a candidate integral does not count as a probability theorem until its density has been derived from the sample.

The named project declarations discussed below have kernel-check evidence. The repository's printed axiom checks use only `propext`, `Classical.choice`, and `Quot.sound` for the audited declarations. This is a dependency statement, not a proof that the mathematical interpretation of every definition matches an external paper. Independent human review of model conventions and analytical specifications remains pending; the scope of the completed model review is specified in Section 6.4.

3 Actual Bartlett coordinates and the Gram density

Let h_1, h_2 be the columns of the central 2×2 sample. Set $S = \|h_1\|^2$. On the event $h_1 \neq 0$, normalize the first column, project the second column onto it, and write the resulting complex coefficient as $Z = A + \mathrm{i}B$. Let E be the squared norm of the orthogonal residual. The implementation gives a measurable unit-direction fallback at zero. It subsequently proves that the exceptional first-column event has mass zero.

Theorem 3.1 (Actual Bartlett law)**.** *The sample-defined coordinates satisfy*

$$\mathcal{L}(S, Z, E) = \mathrm{Gamma}(2, 1) \otimes \mathcal{CN}(0, 1) \otimes \mathrm{Exp}(1),$$

where $\mathrm{Gamma}(\alpha, \beta)$ *uses shape* α *and rate* β*, and* $\mathrm{Exp}(r)$ *uses rate* r*. Relative to* $ds\,da\,db\,de$*, their density is*

$$f_B(s, a, b, e) = \frac{s}{\pi} e^{-(s+a^2+b^2+e)}\, \mathbf{1}_{\{s>0, e>0\}}. \tag{3}$$

Proof. For each fixed encoded first column, the projection and residual calculation gives the same $\mathcal{CN}(0, 1) \otimes \mathrm{Exp}(1)$ fiber law. The direction map is measurable and defined everywhere. The established Gaussian skew-product law transports this fixed-fiber statement to the actual two-column sample, showing independence from the first column. Its energy has the already established $\mathrm{Gamma}(2, 1)$ law. Mapping this joint law to (S, Z, E) gives the displayed product measure. The densities se^{-s}, $\pi^{-1}e^{-(a^2+b^2)}$, and e^{-e} multiply to (3). The positive-domain restriction is derived from these actual marginal laws, including the absence of boundary mass. □

This step is stronger than establishing independence for a fixed projection. The first-column direction is random, so measurability and the invariant fiber law must be used explicitly. The corresponding principal declarations are `actualTwoRow_BartlettCoordinates_law` and `actualTwoRow_BartlettRealCoordinates_eq_normalizedDensity`.

The Gram coordinate chart is

$$T(s, a, b, e) = (s, \sqrt{s}\, a, \sqrt{s}\, b, a^2 + b^2 + e). \tag{4}$$

On the positive Bartlett support $s > 0, e > 0$, its image is D. An auxiliary change-of-variables identity is stated on the larger domain $s > 0$; the actual-density application restricts it to this almost-sure positive support. The inverse on D is

$$(u, b, c, v) \longmapsto \left(u, \frac{b}{\sqrt{u}}, \frac{c}{\sqrt{u}}, v - \frac{b^2 + c^2}{u}\right).$$

The formal chart is proved to reproduce the actual Bartlett Gram mapping. It is not an unrelated parameterization with an assumed connection to H.

Proposition 3.2 (Jacobian and actual Gram density)**.** *On the positive Bartlett domain,* $|\det DT| = s$*. The coordinate law of* H^*H *has density*

$$f_W(u,b,c,v) = \pi^{-1}e^{-(u+v)}\mathbf{1}_D(u,b,c,v). \tag{5}$$

Its integral is one.

Proof. The derivative of (4), in the stated coordinate order, is

$$\begin{pmatrix} 1 & 0 & 0 & 0 \\ a/(2\sqrt{s}) & \sqrt{s} & 0 & 0 \\ b/(2\sqrt{s}) & 0 & \sqrt{s} & 0 \\ 0 & 2a & 2b & 1 \end{pmatrix},$$

whose determinant is s. Lean verifies the Fréchet derivative, the determinant, the inverse, and the image of the domain. The change-of-variables theorem therefore divides the Bartlett density by s. The factor s in (3) cancels, and the exponent becomes $-(u+v)$. The actual-coordinate pushforward and its almost-sure positive support identify the resulting measure with the actual Gram law. Normalization follows from that law being a probability measure; it is not an additional density premise. □

The main declaration is `actualTwoRow_GramRealCoordinates_eq_traceDensity`; the existing Wishart-measure API is exposed by `complexWishart_twoRow_realCoordinates_eq_traceDensity`. Product-measure transport, complex-to-real volume normalization, Gram-coordinate orientation, and the inverse Jacobian are included in the review scope and human review obligations described in Section 6.4.

4 From the positive cone to the minimum law

For a Hermitian matrix W,

$$\lambda_{\min}(W) > x \quad \Longleftrightarrow \quad W - xI \succ 0. \tag{6}$$

The project proves this spectral criterion and its concrete four-coordinate version. It uses the smallest coordinate of the decreasing ordered spectrum.

Theorem 4.1 (Central complex 2×2 minimum)**.** *For the actual sample* (1) *with* $M = 0$ *and* $s = t = 2$*, and* $x \geq 0$*,*

$$\mathbb{P}\{\lambda_{\min}(H^*H) > x\} = e^{-2x}, \tag{7}$$

$$\mathbb{P}\{\lambda_{\min}(H^*H) \leq x\} = 1 - e^{-2x}. \tag{8}$$

For $x < 0$ *the CDF is zero. The complete minimum law is* Exp(2)*.*

Proof. By (6) and Proposition 3.2, the tail is the integral of $\pi^{-1}e^{-\operatorname{tr} W}$ over $W \succ xI$. For $x \geq 0$ this region lies in the positive definite support. Substitute $W = V + xI$. Translation preserves the four-real-coordinate measure, while $\operatorname{tr}(V + xI) = \operatorname{tr} V + 2x$. Hence the integral is

$$e^{-2x}\int_{V\succ 0} \pi^{-1}e^{-\operatorname{tr} V}\, dV = e^{-2x},$$

using the normalization already derived from the actual Gram law. The complement of the strict tail event is exactly the weak CDF event; no exchange of strict and weak inequalities is needed here. Positive semidefiniteness of a Gram matrix gives the negative-threshold case. Finally, uniqueness of probability measures with a common CDF identifies the whole pushforward measure with Exp(2). □

The three principal statements are `actualTwoRow_minimum_tail`, `actualTwoRow_minimum_CDF`, and `actualTwoRow_minimum_law_expTwo`. The unmodified determinant candidate of Jin et al. at $(s, t, L) = (2, 2, 0)$ is also shown equal to this actual CDF in `theorem1CentralTwoRow_candidate_eq_actualCDF`. In this specialization its matrix is explicitly

$$\Psi(x) = e^{-x} \begin{pmatrix} x^2 + 2x + 2 & x + 1 \\ x + 1 & 1 \end{pmatrix}, \qquad \det \Psi(x) = e^{-2x}, \qquad \det \Psi(0) = 1.$$

The Lean candidate takes norms of complex determinants. Here both determinants are positive real numbers, so its norm ratio equals the ordinary determinant ratio in the source formula. This observation is specific to the verified specialization, not a general sign assertion. This last connection is a specialization of their formula, not a proof of their general ordered eigenvalue distribution theorem. The transcription follows arXiv v2 [2]. The journal metadata in [1] has been verified; an equation-by-equation journal comparison is not claimed.

4.1 Positive scalar covariance

Corollary 4.2 (Scale convention)**.** *Let $a > 0$, $v = a^2$, and $W_a = (aH)^*(aH)$. Then*

$$\lambda_{\min}(W_a) = a^2 \lambda_{\min}(H^* H), \qquad \mathcal{L}(\lambda_{\min}(W_a)) = \mathrm{Exp}(2/a^2).$$

In particular, the CDF for variance $v > 0$ is $1 - e^{-2x/v}$ for $x \geq 0$.

Proof. The actual encoded sample matrix respects real scalar multiplication. Consequently its Gram matrix scales by a^2. The spectral threshold criterion identifies the corresponding minimum scaling, and transporting the events in Theorem 4.1 gives the CDF. CDF uniqueness again gives the full law. □

Here a is an amplitude and v is an entry variance. The result addresses a positive scalar covariance; it does not address arbitrary correlated columns.

Remark 4.3 (Monotone log-rate application)**.** *For $a, \rho > 0$, define the nats-per-mode statistic $C_{\min} = \log(1 + \rho \lambda_{\min}((aH)^*(aH)))$. Then, for $R \geq 0$,*

$$\mathbb{P}\{C_{\min}(H; a, \rho) \leq R\} = 1 - \exp\left(-\frac{2(e^R - 1)}{\rho a^2}\right). \tag{9}$$

For $R < 0$ the probability is zero. This follows immediately from $\log(1+\rho g) \leq R \iff g \leq (e^R - 1)/\rho$ for $g \geq 0$, followed by the scaled CDF. The declaration is `actualTwoRow_weakestLogRate_outage`. This application is not counted as a separate principal contribution. A target in bits uses $R \log 2$; coding theorems, power allocation, and total MIMO mutual information are outside the formal statement [12]. The operational interpretation of ρ remains subject to human review.

5 Probability specializations and supporting results

5.1 Fixed orthogonal Gaussian decompositions

Let G have the standard Gaussian law on a finite-dimensional real Hilbert space E, and let K be a fixed linear subspace. Write P_K and $P_{K^\perp}$ for the subspace-valued orthogonal projections.

Proposition 5.1 (Actual orthogonal split)**.** *The projected variables have their subspace standard Gaussian laws and satisfy*

$$\mathcal{L}(P_K G, P_{K^\perp} G) = \gamma_K \otimes \gamma_{K^\perp}.$$

If $\dim_{\mathbb{R}} K = 2k$ *with* $k > 0$*, then* $\|P_K G\|^2/2 \sim \mathrm{Gamma}(k, 1)$*.*

Proof. The characteristic function of a linear image is obtained by composing with the adjoint. For a coisometry the adjoint preserves norm, so the standard Gaussian characteristic function is unchanged. This identifies the projected laws. Joint Gaussianity follows by linear transport, and the cross covariance vanishes by orthogonality. The existing Gaussian independence theorem then gives independence and the product law. The norm statement transports the established Euclidean radial Gamma law through an orthonormal coordinate isometry; it does not require a new radial integration proof. □

These interfaces are `stdGaussian_orthogonalProjectionOnto_law`, `stdGaussian_orthogonal_split_law`, and `stdGaussian_projected_half_norm_sq_law`. They reuse mathlib's Gaussian characterization and independence machinery. In particular, `HasGaussianLaw.indepFun_of_covariance_inner` and `IndepFun.map_prod_eq_prod_map_map` already supply the independence and product-law steps; our code verifies their hypotheses for these projections. The word *fixed* is essential: substituting a random earlier-column subspace requires a separate measurable conditional-law argument.

5.2 Noncentral one-column small-threshold behavior

Theorem 5.2 (Arbitrary positive row count)**.** *For an integer* $m > 0$*, a deterministic* $M \in \mathbb{C}^{m\times 1}$*, and the sample convention* (1)*, let* $F_M(x) = \mathbb{P}\{\|M + H\|_F^2 \le x\}$*. Then*

$$\lim_{x\downarrow 0} \frac{F_M(x)}{x^m} = \frac{e^{-\|M\|_F^2}}{m!}. \tag{10}$$

Proof. Let μ be the real encoded mean, so $\|\mu\|^2 = 2\|M\|_F^2$. The actual event is a real Gaussian ball of radius $r = \sqrt{2x}$. For its centered density φ, translation gives

$$\varphi(z - \mu) = e^{-\|\mu\|^2/2 + \langle z,\mu\rangle}\varphi(z).$$

Cauchy–Schwarz bounds the inner product on that ball by $\pm r\|\mu\|$. Integrating yields

$$e^{-\|M\|_F^2 - r\|\mu\|} F_0(x) \le F_M(x) \le e^{-\|M\|_F^2 + r\|\mu\|} F_0(x).$$

The central energy has the already proved $\mathrm{Gamma}(m, 1)$ law, with $F_0(x)/x^m \to 1/m!$. Dividing the inequalities by x^m and applying the squeeze theorem proves (10). □

The theorem is `noncentralOneColumnSmallestCDF_div_tendsto_frobenius`. It includes zero mean and does not assume a noncentral spectral density. Exact actual-model formula connections for one-column row counts one, two, and three are also present, with the noncentrality parameter matched to the Frobenius mean norm. In particular, the supporting candidate-to-actual-CDF connections include $(s, t, L) = (1, 2, 1)$ and $(1, 3, 1)$, with positive noncentrality and the stated mean-parameter compatibility hypotheses; zero mean is treated by the central branches. These are supporting coverage, not a general multi-column noncentral result.

6 Workflow, verification, and reproducibility

6.1 Declaration-level evidence

The pinned toolchain is Lean `v4.35.0-rc2`; the mathlib revision is `e18afff334217588afc719a8b707b00d90521064`. The repository's `lean-toolchain` and `lake-manifest.json` fix these dependencies. No toolchain upgrade is part of the reported development. Table 1 gives module entry points; the companion claims manifest records exact declarations and source hashes.

Result	Principal source module(s)
Actual coordinate density	`TwoRowActualDensity`; `TwoRowActualNormalizedDensity`
Frobenius frame normalization	`HermitianCoordinateMetric`
Spectral tail and full CDF	`TwoRowMinimumLaw`
Original central candidate	`Theorem1CentralTwoRowActualCDF`
Positive scalar covariance	`TwoRowScaledMinimum`
Log-rate outage	`TwoRowLogRateOutage`
Projection and independent split	`GaussianOrthogonalProjectionLaw`; `GaussianOrthogonalSplitLaw`
Noncentral one-column asymptotic	`T4NoncentralOneColumnAnyRows`
General coordinate cone integral	`CentralHermitianCoordinates`; `CentralWishartConeIntegral`

Table 1: Entry points for the reported statements. The last row is a candidate-kernel identity, not an actual general Wishart law.

Three verification levels are kept distinct: elaborating a leaf module, elaborating the aggregate project entry point, and rebuilding all project sources into fresh output. A cached entry-point check is not a clean rebuild. The publication tooling records fresh-source re-elaboration separately from a full Lake build and does not claim to rebuild mathlib from source. All 164 project modules, counting generated submodules and the aggregate entry, passed this fresh-output source check. One stale theorem identifier in an unreferenced four-row support file was repaired without changing its statement; the interrupted run was then resumed using only unchanged successful outputs. This is one completed source-check campaign, not two independent reproductions. The original report's top-level 74-second interval described only the resumed batch: its initial campaign began earlier, and most successful source outputs were reused within that campaign. No individual historical durations were recorded, so they cannot be reconstructed as measured compilation times. The revision performs a separate fresh-output pass with per-module UTC start and finish times, monotonic wall durations, and explicit batch and campaign origins. Wall durations include helper startup and lock waiting, not just Lean CPU time. The executable status is in the accompanying verification report. That pass checked 164 modules without reuse in 1446.21 seconds. The subsequent acceptance batch adds the frame-normalization module and rechecks the changed aggregate entry, while validating the source and output hashes of reused modules. The resulting snapshot contains 165 modules. This incremental acceptance batch is not another independent full-source reproduction.

The source audit removes nested comments and string literals before searching for prohibited proof placeholders or new axiom declarations. Lexical absence alone is insufficient: declaration-level axiom dependencies and reproducible kernel checking provide the stronger evidence. Independent semantic review is an additional obligation, particularly for the correspondence between formal objects and the intended probabilistic model. The audit enumerates all public theorem declarations in every module in Table 1, including the Wishart API wrapper and both general-coordinate modules. Its binding record contains the source/dependency snapshot hash, the exact audit-source

and output-log hashes, compiled-output hashes, the toolchain/manifest hashes, and the manuscript hash. Changing any source or the manuscript invalidates this binding until it is regenerated and checked.

6.2 Numerical controls

Numerical calculations are used to reject specification mistakes before formal proof work, and as controls for the computational interpretations. They are not proof substitutes. The new log-rate preflight uses seed 20261015 and 200000 actual independent complex Gaussian 2×2 samples. It checks the trace against Gamma$(4, 1)$, the minimum against Exp(2), and the Hermitian quadratic eigenvalue formula against a numerical eigensolver. The corresponding Kolmogorov–Smirnov statistics are approximately 0.00238 for trace–Gamma$(4, 1)$ and 0.00124 for minimum–Exp(2); the maximum eigenvalue-formula discrepancy is below $5 \cdot 10^{-15}$.

Fifteen combinations of amplitude, per-mode coefficient, and target rate verify the event transformation and compare its empirical probability with (9). The reported tolerance is eight binomial standard errors plus $8/N$, not a formal simultaneous confidence guarantee. The script and its JSON output preserve the seed, model, controls, and individual results.

6.3 Recorded AI-assisted workflow and proof-body provenance

The development used Codex for proof decomposition, implementation, numerical controls, and drafting. The local Ollama model `deepseek-prover-v2:7b` was used as a bounded proof-completion tool for short, precisely stated lemmas. Accepted generated proof bodies retain their provenance and the label `GUIDED_COLLABORATION` where applicable.

One archived session (11 October 2026, round three) records 38 runner generations: two passed Lean, 29 were rejected by Lean, and seven were rejected before checking for unusable output. The two accepted bodies establish a normalization coefficient and a squared-norm identity, with targets and assets supplied by Codex; they are not independent density derivations. Two additional manual authoring calls failed and are excluded from the 38-call denominator. Five separate P4-style review requests produced zero usable issue reports. They are not counted as accepted proofs or completed review. These are observational counts from one adaptively guided session, not a controlled benchmark or an estimate of general model accuracy. The runner also stopped on a task-template error; the counts do not measure a full hour of throughput. The artifact includes the event log, actual review requests/responses, and their hashes rather than reporting only successful examples. The rejected review outputs include an end-of-file response, a bare `#print.`, and expository plans instead of issue reports. Thus proof completion and review compliance are separate acceptance criteria: a prover can supply a useful term without being a suitable adversarial reviewer.

Here "Codex" describes the immediate proof body; it does not erase local prover bodies in its dependencies. Conversely, a local helper in the graph does not make the complete density proof a locally generated result.

6.4 Independent review boundary

The failed local-prover review requests are distinct from a subsequent cross-family semantic review by WorkBuddy (Step model family). Its existing report inspected the central 2×2 chain, supporting two- and three-row one-column statements, scaling and outage, fixed-subspace interfaces, the one-column asymptotic, and Frobenius normalization. Within its inspected scope it reported no concrete mathematical errors or hidden assumptions. Its documentary suggestions about chart domains, supporting specializations, and the general-dimensional volume-factor derivation are incorporated

Declaration or family	Direct body	Guidance/upstream composition
`deepseek_scaled_sampleMatrix`	Local DS	Codex target/assets; guided
`deepseek_minimum_DeterminantFactor`	Local DS	Codex target/assets; guided
`deepseek_round3_ComplexScaledCoefficient`; `deepseek_round3_ComplexScaledNormSquare`	Local DS	Codex decomposition; guided
`actualTwoRow_GramRealCoordinates_eq_traceDensity`	Codex	Reuses guided DS normalization helpers
`actualTwoRow_minimum_law_expTwo`	Codex	Actual-density dependency chain
`theorem1CentralTwoRow_candidate_eq_actualCDF`	Codex	Reuses DS determinant body
`actualTwoRow_scaled_minimum_law`	Codex	Reuses DS sample-scaling body
`actualTwoRow_weakestLogRate_outage`	Codex	Reuses actual scaled CDF

Table 2: Proof-body provenance for selected declarations in the headline chain. DS denotes the local prover tool, not an academic author. Source hashes and the transitive helper list are in the artifact.

here. The report, reviewed snapshot hashes, and an issue-disposition record accompany the artifact. The Lean sources are unchanged from that reviewed snapshot; the present author, metadata, and editorial updates are separately recorded.

This was a scoped model review, not comprehensive human mathematical review. The report lists 32 files read in full, two auxiliary files checked only at their statement signatures, and approximately 130 project modules plus mathlib outside its proof-body review. In particular, full radial/Bessel dependencies, the general cone-integral implementation, and general source-formula branches are not covered by that report. Human obligations remain: checking measurable skew-product and Fubini specifications, complex-to-real and intrinsic Frobenius volume interpretations, sample variance and Gram orientation, strict/weak spectral events, and formal-definition correspondence to source notation. No general journal equation comparison or complete semantic acceptance follows from the absence of findings in the inspected subset.

AI tools are not authors. The named human author must verify and take responsibility for the mathematical and bibliographic content, consistently with arXiv's tool-disclosure policy [13].

7 Generalization boundary and research direction

For $0 < s \le t$, the desired actual central Gram density relative to a fixed s^2-real-coordinate Hermitian volume is

$$K_{s,t} \det(W)^{t-s} e^{-\operatorname{tr} W} \mathbf{1}_{W \succ 0}, \qquad K_{s,t} = \left(\pi^{s(s-1)/2} \prod_{j=1}^{s} (t-j)! \right)^{-1}. \tag{11}$$

This is the coordinate-volume coefficient. Relative to induced Frobenius volume it becomes $2^{-s(s-1)/2} K_{s,t}$; convention differences must be resolved before comparing a coefficient with an external source. The project has *not* derived (11) from the actual Gaussian sample in arbitrary dimension. What is formalized is the translation identity for the corresponding candidate-kernel integral. If the actual density and normalization are subsequently established, the translation would give, for $x \ge 0$,

$$\mathbb{P}\{\lambda_{\min} > x\} = e^{-sx} K_{s,t} \int_{V \succ 0} \det(V + xI)^{t-s} e^{-\operatorname{tr} V} \, dV.$$

The determinant factor must be retained when $t > s$. The kernel identity itself can be stated for negative x; its interpretation as a tail probability then requires the original positive-support indicator and cannot simply be reused unchanged.

The next substantive step is a general Cholesky density induction, rather than a collection of further low-dimensional charts. Four obligations are explicit. First, measurable parameterized projections must lift the fixed-subspace laws to the random earlier-column conditional setting. Second, for $r > 0$ and $z \in \mathbb{C}^{s-1}$ the block map

$$(r, z, B) \longmapsto \begin{pmatrix} r & \sqrt{r}\, z^* \\ \sqrt{r}\, z & B + zz^* \end{pmatrix}$$

must have a dimension-independent real-coordinate absolute Jacobian r^{s-1}. Small-dimensional symbolic controls are available but are not a general Lean proof. Third, the determinant and trace identities must combine with this Jacobian and the recurrence

$$K_{s,t} = \frac{K_{s-1,t-1}}{\pi^{s-1}(t-1)!}.$$

Finally, exceptional sets and full-rank support must be connected to the actual pushforward measure. These obligations identify reusable probability and multivariate-calculus contributions beyond the present example.

General noncentral ordered spectral densities and the full parameter range of the original Theorems 1–4 remain outside the established results. In particular, the present cone method does not supply a noncentral unitary integration formula. The current work supplies a verified example and interfaces for pursuing those questions, rather than a completed formalization of the whole source paper.

8 Conclusion

The development connects real Gaussian samples to an actual complex Gram density and then to a smallest-eigenvalue law, with scalar covariance and logarithmic-rate event transport made explicit. Its main reusable lesson is the separation of actual pushforward laws, coordinate transformations, and candidate formulas. The arbitrary-row one-column asymptotic and fixed orthogonal Gaussian split provide further foundations for a general density induction. The completed cross-family review has a stated scope; final human mathematical signoff, the related-work assessment, and release-rights decisions remain the responsibility of the author.

A Artifact organization and release conditions

The companion repository places Lean sources under `theory/jin_wishart_formalization/`, numerical controls under `experiments/`, and exact claims and verification reports under `verification/`. This manuscript is self-contained LaTeX; its bibliography is included below, with no external images, custom class, or local input files. The files `claims.json`, `REPRODUCIBILITY.md`, and `SUBMISSION_READINESS.md` accompany the working manuscript.

The submission source layout includes a code archive under `anc/wishart_formalization_code.zip`, preserving the project paths, exact dependency pins, claims, and verification evidence. This local preparation does not itself publish the artifact. Before submission, the author must confirm release rights and the applicable license, verify the final arXiv output, and resolve or explicitly

narrow the remaining human review items. Fresh-source evidence must be checked against the packaged source snapshot. A final related-work search must support the precise contribution claims. Internal normalization diagnostics are excluded from this manuscript pending source reconciliation and identifiable parameter tests. arXiv moderation is not peer review [13]; formatting compliance does not imply acceptance.

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